Highlights

Prune Bias From the Root: Bias Removal and Fairness Estimation by Pruning Sensitive Attributes in Pre-trained DNN Models

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- **Prove the effectiveness of single-attribute pruning for bias removal on DNNs**.
- **Propose the use of attribute pruning on multiple features**.
- **Introduce a robust multi-attribute fairness estimation method**.
- **Explicitly discuss the limits of algorithmic bias removal and provide suggestions**.

Prune Bias From the Root: Bias Removal and Fairness Estimation by Pruning Sensitive Attributes in Pre-trained DNN Models

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ABSTRACT

Context: Deep learning models (DNNs) are widely used in high-stakes decision-making domains, but they often inherit and amplify biases present in training data, leading to unfair predictions. Given this context, fairness estimation metrics and bias removal methods are required to select and enhance fair models. However, we found that existing metrics lack robustness in estimating multi-attribute group fairness. Further, existing post-processing bias removal methods often focus on group fairness and fail to address individual fairness or optimize along multiple sensitive attributes.
Objectives: In this study, we explore the effectiveness of attribute pruning (i.e., zeroing out sensitive attribute weights in a pre-trained DNN's input layer) in both bias removal and multi-attribute group fairness estimation.
Methods: To study attribute pruning's impact on bias removal, we conducted experiments on 32 models and 4 widely used datasets, and compared its effect in single-attribute group bias removal and accuracy preservation with 3 baseline post-processing methods. We then leveraged 3 datasets with multiple sensitive attributes to demonstrate how to use attribute pruning for multi-attribute group fairness estimation.
Results: Single-attribute pruning can better preserve model accuracy than conventional post-processing methods in 23 out of 32 cases, and enforces individual fairness by design. However, since individual fairness and group fairness are fundamentally different objectives, attribute pruning's effect on group fairness metrics is often inconsistent. We also extend our approach to a multi-attribute setting, demonstrating its potential for improving individual fairness jointly across sensitive attributes and for enabling multi-attribute fairness-aware model selection.
Conclusion: Attribute pruning is a practical post-processing approach for enforcing individual fairness, with limited and data-dependent impact on group fairness. These limitations reflect the inherent trade-off between individual and group fairness objectives. In addition, attribute pruning provides a useful mechanism for bias estimation, particularly in multi-attribute contexts. We advocate for its adoption as a comparison baseline in fairness-aware AI development and encourage further exploration.

1. introduction

Given their vital power, machine learning models such as deep learning models (DNNs) are widely applied in high-stakes domains such as finance, healthcare, and law due to their exceptional predictive power. Ideally, these models should be bias-free and provide fair, objective predictions to aid practitioners in making decisions [1, 2]. However, in real-world scenarios, DNNs often inherit and amplify historical biases present in training data, leading to ethically problematic and unreliable outcomes [3, 4]. Trained on biased datasets, these models can make predictions based on sensitive attributes such as race, gender, and age, resulting in discriminatory decisions that reinforce societal inequalities. For example, in the widely studied COMPAS dataset [5] for recidivism prediction, DNN models have been shown to exhibit racial bias, leading to disproportionately unfavorable predictions for black individuals compared to white individuals. Such bias has a significant negative social impact, thus, it is necessary to adopt fairness evaluation metrics and bias removal methods in machine learning systems.

To address machine learning bias, researchers have developed various fairness metrics, broadly categorized into **group fairness** and **individual fairness**. Group fairness metrics [6, 7, 8, 9] focus on ensuring equitable treatment across predefined demographic subgroups. On the other hand, individual fairness aims to ensure that similar individuals receive similar outcomes, independent of their sensitive attributes [10, 11, 12, 7, 13]. This finer-grained approach aligns more closely with the principle of fairness in personalized decision-making [14]. Both metrics are important in

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understanding the fairness of machine learning models and can aid practitioners in fair model selection regarding different actual needs. However, we found that existing group fairness metrics often focus on a single sensitive attribute [6, 7], and metrics addressing multiple sensitive attributes risk of experiencing overfitting on small subgroups, whose observed distribution may fail to represent the real distribution [8, 9].

To mitigate bias in machine learning, fairness-aware interventions have been developed at different stages of the modeling pipeline. These approaches can be categorized into **pre-processing**, **in-processing**, and **post-processing** methods. Pre-processing techniques adjust training data distributions to reduce bias before model training [15, 16, 17, 18, 19, 20, 21, 22, 23]. In-processing methods incorporate fairness constraints directly into model optimization to suppress biased decision pathways [23, 19, 15, 24, 25, 26, 27, 28]. Post-processing approaches adjust model predictions after training, making them particularly efficient since they do not require retraining [6, 29, 30].

While existing post-processing techniques efficiently and effectively adjust outcome distributions, we noticed that they mostly only focus on group fairness and few address the issue of individual fairness, which limits their applicability when individual fairness is a priority. Another issue worth noticing is that existing post-processing techniques often focus on single sensitive attribute fairness correction, which makes them inapplicable in more complex scenarios when multiple sensitive attributes (e.g., gender and race) affect the prediction results.

As a solution to the above-mentioned issues, we promote the use of attribute pruning in fairness estimation and DNN bias removal. While pruning has been a popular topic in machine learning [31], and attribute pruning has been tested beneficial in improving model fairness on large models such as GPT-2 [32], its generalizability on small-scale DNNs has not yet been studied. Moreover, this approach has not yet been used for fairness estimation purposes. By directly pruning (i.e., zeroing out) the weights associated with sensitive attributes in the input layer of a trained DNN, we can enforce strict individual fairness (i.e., predictions remain independent of sensitive attributes) without modifying deeper layers of the model or requiring additional retraining. In this study, we reveal the effectiveness of sensitive attribute pruning on small-scale DNN bias removal and discuss its usage in multi-attribute fairness estimation by answering the following research questions:

RQ1: How does single-attribute pruning perform in comparison to the existing post-processing methods?
By answering this research question, we aim to understand the accuracy and group fairness impact of single-attribute pruning on 32 models and compare them with 3 state-of-the-art post-processing methods.

RQ2: How does multi-attribute pruning impact and aid understanding of the original models?
By answering this research question, we investigate the accuracy impact of multi-attribute pruning on 24 models. Further, we investigate the prediction change brought by attribute pruning on different subgroups and discuss their implications on multi-attribute fairness estimation.

According to our experiments on 32 models and 4 most widely used datasets, we found that single sensitive attribute pruning can effectively remove bias while maintaining the model's original accuracy. It preserved the **highest post-correction accuracy** in 23 out of 32 datasets, outperforming traditional post-processing methods such as equalized odds, calibrated equalized odds, and reject option classification. However, attribute pruning does not always optimize group fairness, which confirms the fundamental trade-off between individual fairness and group fairness addressed in a recent study by Binns [14].

Multi-attribute pruning has a stronger accuracy impact than single-attribute pruning, yet the accuracies are still higher than conventional single-attribute post-processing bias removal methods in most cases. Further, by analyzing true positive rate (TPR) differences across subgroups, one can robustly quantify the influence of sensitive attributes on model prediction, which enables fairness-aware model selection. Practitioners can compare subgroup TPR differences to select models based on their fairness objectives, ensuring either general fairness or targeted fairness improvements for specific subgroups.

Motivated by the results obtained along our research questions, we dive further into the implications revealed in the results and justify the limitations of algorithmic bias removal approaches. Through our experiments and discussions, we promote the usage of attribute pruning in bias removal and fairness estimation. Our experiment data is available at: `https://zenodo.org/records/16542215`.

To sum up, our paper makes the following contributions:

1. Assess the effectiveness of post-processing single-attribute pruning on smaller-scale DNNs using 32 models with different network structures.
2. Assess the accuracy impact of post-processing multi-attributes pruning and promote its usage for individual fairness protection under complex scenarios.

3. Propose a highly informative multi-attribute fairness estimation approach based on attribute pruning, which is robust to subgroup distribution distortion.

The remainder of our study is organized as follows. We first introduce the reach background in machine learning fairness and motivate our study on DNN sensitive attribute pruning in Sec. 2. We then introduce our experiment in Sec. 3, including datasets, models, benchmarks, evaluation and metrics. The results are discussed in Sec. 4. Based on our results, we discuss the implications of our results and address the limitations of algorithmic bias removals in Sec. 5. Sec 6 discusses the validity threats we may encounter in the research. Finally, we provide the conclusions on our findings in Sec 7.

2. Background and Motivation

2.1. ML Bias and Fairness

Machine learning models learn to extract and represent features using datasets and are highly effective. Given their good performance, machine learning models such as DNN have been applied in different fields, such as finance, law, and healthcare [33]. However, due to historical reasons, bias on sensitive attributes (e.g., sex, race, age) may occur in a dataset and therefore harm the fairness of the model. For example, the COMPAS dataset used for recidivism of pretrial offenders prediction favors white people compared to black people [5]. The model trained on this dataset will thus embed the bias in race and provide unfair judgments based on this sensitive attribute. These judgments could lead to negative impacts on society, and therefore, the bias in machine learning models needs to be identified and mitigated.

Different metrics have been proposed to quantify bias and fairness in machine learning. These fairness metrics can be categorized into **group fairness** and **individual fairness**. Group fairness metrics such as demographic parity (DP) [7] and equal opportunity (EO) [6] measure the fairness at the binary sensitive subgroup level (e.g., male vs. female, white vs. non-white). For example, under a binary-classification task, demographic parity regards the model to be fair if the ratios of positively predicted individuals in different subgroups are the same (e.g., the model is regarded as fair on the sex attribute if both the female and male subgroups receive 30% positive prediction outcomes). However, most of the group fairness metrics are coarse-grained, which focus on only one sensitive attribute with two value choices. To generalize the estimations on multiple attributes, researchers proposed to investigate fairness on different combinations of sensitive attributes using positive rate or true positive rate calibrations. The problem with this approach is that the subgroup may be overly fine-grained, and the observed samples cannot reflect the real distributions. For instance, an Inuit female individual has a much lower probability of existing in a bank marketing dataset than a white male individual. Hence, the group fairness calculated and enhanced on these subgroups may lead to overfitting [8, 9], and provide a distorted estimation of bias.

On the other hand, individual fairness metrics such as counterfactual fairness [11] and disparate treatment [10, 13] investigate fairness on a finer granularity, and consider the model is fair if similar individuals are receiving the same outcome. A desired model satisfying this requirement should output the same prediction for individuals with the same non-sensitive attributes, regardless of their sensitive attributes. For instance, a strictly fair DNN model should predict female and male individuals to have an identical likelihood (e.g., softmax score) of belonging to high revenue if their educational and working experiences are the same; a relaxed constraint is that the model should have the same decision, but not necessarily the identical likelihood on similar individuals mentioned above [7]. However, if a model only satisfies the relaxed version of fairness (i.e., ensuring the same decision rather than identical probability), small variations in inputs may lead to different probabilities that cross a decision threshold due to Lipschitz continuity in predictions [7]. To minimize the margin of decision flip in the whole input space, the likelihood difference of each sample pair shall be suppressed below a certain small threshold, which can be considered "unbiased". Given that the model is unbiased under a certain threshold for all inputs, which decisions are regarded as independent of the sensitive attributes, each neuron (i.e., the internal representations of a model) should ideally exhibit similar distributions for individuals who are alike except for their sensitive attributes. This ensures that sensitive information does not implicitly drive decisions at deeper network layers. A toy example of biased and unbiased neuron outputs is shown in Fig.1.

The notion of individual fairness is motivated by the "unfairness" of group fairness metrics on the individual level. For instance, according to demographic parity, a company may be requested to hire 30% male candidates and 30% female candidates. However, forcing such strict group fairness in reality may lead to unfair decisions, such as replacing a more qualified male candidate with a less qualified female candidate [7]. Apart from the unfair treatment of individuals, forcing only group fairness may also lead to a decrease in decision accuracy, as the decisions do not

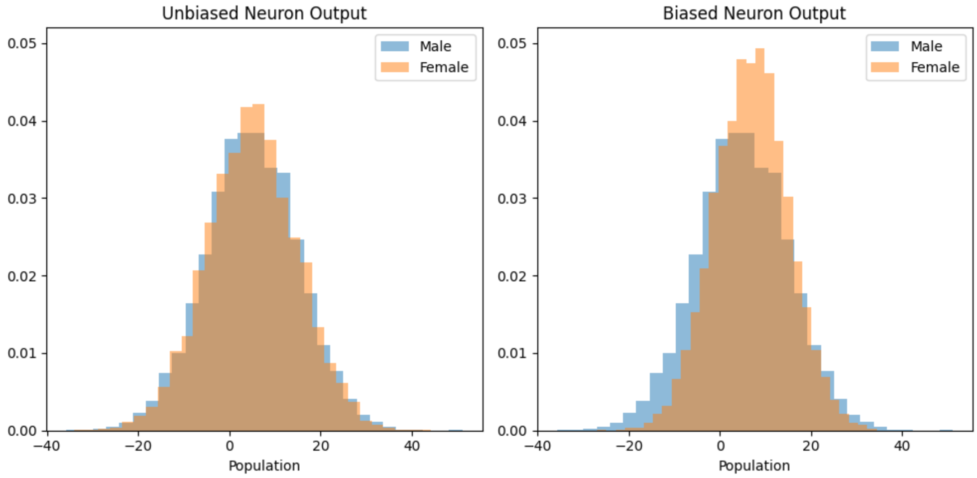

Figure 1: Biased v.s. unbiased neuron output on gender.

fully meet the outcome probabilities learned with labeled datasets [14]. In contrast, considering individual fairness can better ensure the accuracy of decisions while also eliminating the effect of sensitive attributes on the outcomes [7, 12].

Given the different definition of the two types of metrics, Binns [14] argues that there are inherent trade-offs between group fairness and individual fairness: **although individual fairness ensures a more equitable decision, the outcomes can sometimes be seemingly unfair in group fairness, which stresses more on equality**. Therefore, both metrics should be considered to provide a comprehensive understanding of fairness, but their balances and priorities should be considered according to actual requirements.

2.2. Bias Removal Algorithms

As previously introduced, there are three types of bias removal approaches, namely pre-processing [15, 16, 17, 18, 19, 20, 21, 22, 23], in-processing [23, 19, 15, 24, 25, 26, 27, 28], and post-processing [6, 29, 30]. Pre-processing bias removal methods identify biased features, analyze data distributions, and modify the dataset before the model training to prevent a biased learning process [15, 16, 17, 18, 19, 20, 21, 22, 23]. These methods can enhance individual fairness by removing sensitive attributes or adjusting the data distribution. In-processing methods encode the fairness requirements during model training and mitigate the bias through regularization or adversarial training [23, 19, 15, 24, 25, 26, 27, 28]. Moreover, researchers proposed in-processing approaches to discover biased paths and leverage model repairing methods to mitigate the biases in these neurons [34, 35, 36]. While these methods are often highly effective and work exceptionally well for heavily biased models [34], they often require an extra retraining process for parameter optimization, which may be inefficient when the data is large or the model is complex. Furthermore, approaches for pre-processing and in-processing are somehow radically different than those for post-processing. Therefore, initially, we focus on a more efficient category, post-processing bias removal algorithms, in our presented study, leaving the study and comparison with the other two approaches for further research.

In comparison to pre-processing and in-processing methods, post-processing bias removal algorithms work without retraining the models and thus are highly efficient. The three most commonly used post-processing methods are equalized odds [6], calibrated equalized odds [29], and reject option classification [30]. These methods analyze the distribution of samples predicted as positive in each sensitive attribute group and directly adjust the decisions to optimize group fairness metrics such as demographic parity.

Although post-processing methods are feasible to apply, we noticed that there are few studies on post-processing bias removal regarding individual fairness [37]. Further, these methods are limited to single-attribute bias removal based on the prediction outcomes, which limits their applicability in more complex scenarios.

2.3. Post-processing Attribute Pruning

As mentioned in Sec. 2.2, existing post-processing bias removal approaches often neglect the optimization on individual fairness and fail to address the issue of multi-attribute bias removal. This is because individual fairness is difficult to accurately measure on a limited set of observed datasets, while these post-processing bias removal methods engineer the decisions instead of modifying the original, biased model. On the other hand, individual fairness-aware bias removal can be achieved by direct attribute pruning on the first layer.

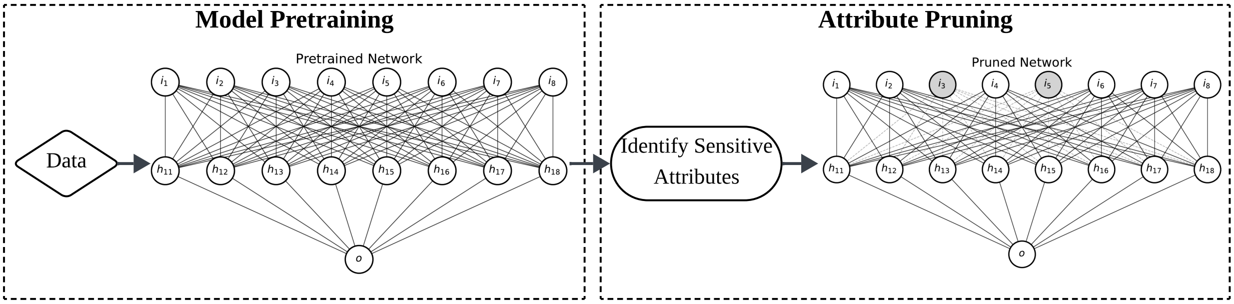

Figure 2: The process overview of post-processing attribute pruning. Identified sensitive attributes are highlighted with gray backgrounds, and the dashed lines in the pruned network indicate weights set to zero. i_n indicates the n^{th} input node, h_{nm} indicates the m^{th} node in the n^{th} hidden layer, and o indicates the output node.

We illustrate the process of attribute pruning with a simple DNN example in Fig. 2. Sensitive attributes such as sex and age are first identified before practicing the pruning process. These weights for the identified sensitive input nodes) **are zeroed out in the first DNN layer** in the pretrained network. This action ensures that the pruned attribute will not propagate to connected neurons in the deeper layers. By cutting the direct correlations between sensitive and non-sensitive attributes, the similar individuals who have differences only in the predefined sensitive attributes will receive the same prediction probability.

The approach shares similarity with traditional model pruning. Model pruning stands for the process of reducing or disabling the neurons that do not significantly contribute to the model's decision. Pruning can take place in different stages of model training. One of the most popular pruning categories is "pruning after training", where the action takes place on a pre-trained model [31]. Pruning after training often follows two schemas: **Pretrain-Prune-Retrain** [38, 39] and **Pretrain-Prune** [40, 41]. The first schema requires retraining or finetuning after the pruning process to adjust the correlations between the remaining neurons. However, studies on the second schema have proven that skipping the retraining process will not lead to significant performance drops, especially on large models.

The idea of post-processing attribute pruning bias removal can be regarded as post-training unstructured pruning, with the Pretrain-Prune schema applied. The difference between our approach and traditional model pruning is that we only modify the neurons in the first layer, and the neurons explicitly correspond to specific input attributes. The removal also disregards the value of the original weight, while only the neurons with negligible weights are pruned in traditional model pruning. Although such similar settings have been proven effective on large models such as GPT-2 [32], their effectiveness has not yet been thoroughly evaluated on small DNN models, where feature signals are often densely mixed and less distinguishable. Given the dense correlation in small-scale DNN, a major reason for not directly using attribute pruning post-processing these models is based on the belief that, given a trained small-scale DNN model with (near-)optimal weights, dropping specific attributes would harm the correlations among these and other attributes already learned by the model, and accuracy may drop dramatically. Another concern towards sensitive attribute pruning is that sensitive information might be encoded redundantly in proxy features that are regarded as non-sensitive attributes (e.g., postal codes indicating races) [42, 33, 43]. In the following paragraph, we analyze why **post-processing attribute pruning should still be applicable with these premises**.

2.3.1. Dense Signal Correlation in Smaller-scale DNNs

Existing post-processing methods require an explicit indication of sensitive attributes, which provides the initial requirement of attribute pruning (i.e., which attributes are to be disabled). Given that the model is assumed to be biased, it will make predictions based on sensitive attributes to some extent (i.e., non-zero weights on the sensitive attribute), and pruning sensitive attributes would lead to a change in accuracy. However, if removing such sensitive attributes severely harms accuracy, then it indicates that the decisions are made mainly with these attributes. Two reasons may lead to this outcome:

1. The dataset is heavily biased in distribution in an undesired way (e.g., the dataset contains only 1% female records in high revenue, but 30% male records fall in this category).
2. The "sensitive attributes" are actually required for the decision (e.g., the "age" attribute when predicting whether an individual has the risk of getting a stroke).

If the outcome is due to the second reason, then the attribute should not be considered as a correction objective, because dropping this attribute can deeply harm the interpretability of the prediction [44, 45]. On the other hand, bear in mind that post-processing bias removal adjusts the outcome distribution based on the original predictions [6, 29, 30]: if the accuracy drop is due to the first reason, then data adjustment and model re-training are much more preferred than any post-processing method [34].

2.3.2. Proxy Features Encoding Sensitive Information

In existing literature, individual fairness is defined as "ensuring similar individuals receive similar prediction outcomes", and the similarity is defined based on a set of chosen attributes [14, 7]. Under this definition, an individually-fair model would output the same predictions for individuals with the same **chosen attributes,** even if their sensitive attributes differ.

Given that similarity is defined solely over the chosen features, an individually-fair model only guarantees equality for individuals with identical selected feature vectors, and proxy features encoding sensitive information are not discussed in the context of individual fairness. Binns notes that mitigating such implicit bias falls under the broader concept of individual justice, which cannot be fully achieved by machine learning models trained on limited data observations [14].

Attribute pruning removes the direct contribution of sensitive attributes to the model's predictions. Two individuals are regarded as similar if they share the same non-sensitive attributes. Since individual fairness does not account for the implicit encoding of sensitive information in non-sensitive attributes, our approach ensures strict individual fairness by design, with respect to the chosen non-sensitive attributes, while acknowledging that implicit biases via proxy features may remain. We extend the discussions on approximating individual justice with attribute pruning in Sec. 5.1.

3. Study Design

3.1. Datasets

To comprehensively understand the impact of sensitive attribute pruning, we select four commonly used fairness datasets for classification collected from different domains [52], namely Bank Marketing (BM), German Credit (GC), Adult Census (AC), and COMPAS. We select the four datasets because they are most widely and frequently used in fairness studies, and provide a wide range of corresponding pre-trained models used in existing research. An overview of the datasets and the model in Table 1. Their training/testing split aligns with the previous studies [53, 51].

The Bank Marketing[1] dataset consists of marketing data from a Portuguese bank, containing 45,222 instances with 16 attributes, and the biased attribute identified is **age**. The objective is to classify whether a client will subscribe to a term deposit.

The German Credit dataset[2] includes 1,000 instances of individuals who have taken credit from a bank, each described by 20 attributes, with two sensitive attributes, **sex and age**; the single sensitive attribute to be evaluated in RQ1 is **age**, given that the subgroup positive rate difference (i.e., historical bias in the label) on this sensitive attribute is higher than **sex**. The task is to classify the credit risk of an individual.

The Adult Census dataset[3] comprises United States census data from 41,188 individuals after empty entry removal, with 13 attributes. The sensitive attributes in the dataset are **sex and race**; the single sensitive attribute to be evaluated in RQ1 is **sex**. The goal is to predict whether an individual earns more than $50,000 per year.

The COMPAS (Correctional Offender Management Profiling for Alternative Sanctions)[4] dataset is collected from a system widely used for criminal recidivism risk prediction, containing 6,172 individuals and 6 attributes. The sensitive attributes in the dataset are **race and age**; to keep aligned with previous research [51], the single sensitive attribute to be evaluated in RQ1 is **race**. The goal is to predict whether an individual will reoffend in the future.

[1] https://archive.ics.uci.edu/dataset/222/bank+marketing
[2] https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data
[3] https://archive.ics.uci.edu/dataset/2/adult
[4] https://mlr3fairness.mlr-org.com/reference/compas.html

Table 1
An overview of the datasets and benchmark model used in the experiments.

Dataset	Introduction	Sens. Attributes & Pos. Rate		Num. Samples	Models	# Neurons
Marketing: Bank Marketing (BM)	Marketing data for client term deposit subscription prediction.	age	age=0: 0.230 age=1: 0.124	41,188	BM1 (Kaggle)	97
					BM2 (Kaggle)	65
					BM3 [46, 47]	117
					BM4 (Kaggle)	318
					BM5 (Kaggle)	49
					BM6 (Kaggle)	35
					BM7 (Kaggle)	145
					BM8 [48]	141
Finance: German Credit (GC)	Bank data for credit risk prediction.	age, sex	age=0: 0.578 age=1: 0.728 sex=0: 0.689 sex=1: 0.695	1,000	GC1 (Kaggle)	64
					GC2 [46, 47]	114
					GC3 (Kaggle)	23
					GC4 (Kaggle)	24
					GC5 [48]	138
Economics: Adult Census (AC)	US census data for revenue prediction (i.e., whether the person earns over $50,000).	sex, race	sex=0: 0.114 sex=1: 0.312 race=0: 0.122 race=1: 0.283 race=2: 0.126 race=3: 0.127 race=4: 0.262	45,222	AC1 (Kaggle)	45
					AC2 [46, 47]	121
					AC3 (Kaggle)	71
					AC4 (Kaggle)	221
					AC5 (Kaggle)	149
					AC6 (Kaggle)	45
					AC7 [48]	145
					AC8 [49, 50]	10
					AC9 [49, 50]	12
					AC10 [49, 50]	20
					AC11 [49, 50]	40
					AC12 [49, 50]	45
Law: COMPAS	Data from the COMPAS (Correctional Offender Management Profiling for Alternative Sanctions), used for scoring a criminal's likelihood of reoffending crimes.	race, age	race=0: 0.113 race=1: 0.333 age=0: 0.140 age=1: 0.305	6,172	Compas1 [51]	24
					Compas2 [51]	124
					Compas3 [51]	600
					Compas4 [51]	90
					Compas5 [51]	2,000
					Compas6 [51]	4,000
					Compas7 [51]	10,000

3.2. Models

The benchmark DNN models for the four datasets are provided by a previous study by Kim *et al.* [51], and their structures are shown in Table 1. The benchmark models were gathered from previous studies [46, 47, 49, 50, 48, 51] and Kaggle. To ensure correct implementation, we obtained the models and their parameters from the replication package provided by Kim *et al.* [51].

3.3. Evaluation Metrics

To evaluate the accuracy and fairness impact of post-processing on the outcome, we evaluate the results using **accuracy**, **demographic parity**, and **equal opportunity** on single-attribute pruning, and asses the change in **accuracy** on multi-attribute pruning. Note that we only evaluated the fairness with group fairness metrics, since individual fairness is often evaluated on the system and is infeasible to assess directly on decisions, where the data used for prediction is a subset observation of the input space. Moreover, according to the analysis in Sec. 2.3, our attribute pruning method can achieve perfect individual fairness on the concerned sensitive attributes, given that the sensitive attributes will not contribute weights to the prediction likelihood.

Demographic Parity (DP) [7]. Demographic parity (i.e., statistical parity) is a widely used metric to evaluate group fairness. It defines fairness as ensuring that subpopulations with different sensitive attributes (e.g., $S = 0$ and $S = 1$) have an equal probability of being classified as positive (i.e., $\hat{Y} = 1$):

$$DP = |P(\hat{Y} = 1|S = 0) - P(\hat{Y} = 1|S = 1)| \tag{1}$$

According to the definition, a smaller value of DP indicates that the decision $\hat{Y}$ is statistically less dependent on the sensitive attribute S.

Equal Opportunity (EO) [6]. Another commonly used fairness metric is equal opportunity, which compares the true positive rates (TPR) between different subpopulations:

$$EO = |P(\hat{Y} = 1|S = 0, Y = 1) - P(\hat{Y} = 1|S = 1, Y = 1)| \tag{2}$$

Equal opportunity considers the potential differences in true positive rates across subgroups. A lower EO value suggests that the likelihood that an individual receives a positive prediction is more independent of the sensitive attribute.

While demographic parity ensures that positive predictions are distributed equally across groups, equal opportunity focuses on achieving fairness specifically for those who deserve positive outcomes. We leverage both metrics in evaluation to provide a more comprehensive understanding of the performance of post-processing bias removal metrics.

To further understand whether there is a significant group fairness difference before and after applying the bias correction algorithm, we also carried out paired t-tests on each dataset for every applied algorithm. The null hypothesis for paired t-tests is that **the two data sequences come from the same distribution** (i.e., no significant difference is observed). If the p-value between the two metrics is smaller than 0.05 (5% significance level), we say there is a significant difference between the metric before and after applying the correction algorithm. Otherwise, we say that the difference is not significant. We also conduct a one-sided normal tolerance interval test to estimate the upper bound (with 95% confidence and 95% population coverage) of the accuracy decrement (i.e., original accuracy-corrected accuracy) before and after applying each post-processing method. An ideal correction algorithm for group fairness should significantly improve the fairness metrics while incurring a minimal decrement in accuracy.

3.4. Experiment Design

3.4.1. Single-attribute Approach

To answer RQ1, we use the metrics in Sec. 3.3 to compare our single-attribute pruning approach (i.e., zeroing out the model weight in the first layer for a single, binary sensitive attribute) with the most common three post-processing methods: equalized odds (EqOdds) [6], calibrated equalized odds (CalEqOdds) [29], and reject option classification (ROC) [30]. The three methods are implemented with the functions provided in AIF360[5]. The parameters remain the default during implementations. All three methods are fitted on the training set and made predictions on the testing set.

Equalized Odds [6]. The equalized odds algorithm adjusts the predicted labels to satisfy the equalized odds criterion. In other words, the method aims to ensure the false positive rate (FPR) and false negative rate (FNR) are the same across different subgroups defined by a single sensitive attribute. The algorithm sets up a linear optimization problem to find the probabilities by which it should perturb the predicted labels. This involves defining decision variables for the probabilities of changing the predicted label under certain conditions (e.g., changing from a positive label to a negative label or vice versa, for both privileged and unprivileged groups).

Calibrated Equalized Odds [29]. The calibrated equalized odds algorithm modifies the decision to satisfy the equalized odds constraint while preserving calibration. This post-processing method introduces a relaxation of Equalized Odds, ensuring that weighted sums of group error rates are matched rather than strictly equalized. The approach involves defining relevant groups, establishing an equal-cost constraint that balances FPR and FNR, and adjusting predictions to minimize discrepancies.

Reject Option Classification (ROC) [30]. The reject option classification algorithm adjusts predictions in regions of high uncertainty (i.e., predictions close to the decision boundary). The method identifies a “reject option” region, a band around the classification threshold where the classifier is least confident. Within this region, the algorithm favors unprivileged groups by assigning them positive outcomes while penalizing privileged groups by assigning them negative outcomes. The algorithm optimizes a given fairness metric (e.g., demographic parity, average odds difference) by searching over a range of classification thresholds and ROC margins. Given the modifications are expected to be close to the decision boundary, the method is expected to preserve the original level of classification accuracy. Following the default settings in AIF360, the fairness optimization target used in our experiments is demographic parity, with upper and lower bounds of 0.05 and -0.05, respectively; the lowest and the highest classification thresholds for parameter searching are 0.01 and 0.99.

[5]https://github.com/Trusted-AI/AIF360

Table 2
The original performance (i.e., accuracy, demographic parity, and equal opportunity) of each model and the performance after each post-processing correction. The best performances (i.e., highest accuracy, lowest DP and EO) are highlighted in bold, and the worst performances are marked in italics.

Dataset	Models	Post-Processing Method														
		Original			Attribute Pruning			EqOdds			CalibratedEqOdds			ROC		
		Acc	DP	EO	Acc	DP	EO	Acc	DP	EO	Acc	DP	EO	Acc	DP	EO
Bank Marketing (BM)	BM1	0.892	0.012	0.164	**0.894**	0.010	0.177	0.883	**0.001**	0.175	0.892	0.010	*0.229*	*0.882*	*0.016*	**0.000**
	BM2	0.888	0.007	0.182	**0.887**	**0.005**	0.187	0.862	0.012	0.155	**0.887**	*0.023*	*0.279*	*0.882*	0.015	**0.000**
	BM3	0.882	0.032	0.156	0.878	*0.021*	0.184	0.848	0.013	0.204	**0.882**	0.010	*0.220*	*0.869*	**0.000**	**0.000**
	BM4	0.895	0.006	0.197	0.893	**0.000**	0.211	0.885	0.013	0.233	**0.895**	*0.024*	*0.294*	*0.868*	0.006	**0.004**
	BM5	0.889	0.009	0.164	0.888	0.005	0.176	0.859	*0.031*	0.205	**0.889**	0.013	*0.229*	*0.869*	**0.000**	**0.000**
	BM6	0.889	0.000	0.189	**0.890**	**0.007**	0.219	0.875	0.026	0.202	0.889	*0.029*	*0.286*	*0.881*	0.018	**0.015**
	BM7	0.887	0.008	0.165	**0.888**	**0.004**	0.184	0.860	*0.030*	0.209	0.886	0.029	*0.294*	*0.881*	0.017	**0.021**
	BM8	0.892	0.007	0.128	0.891	**0.001**	0.146	0.884	0.012	0.142	**0.892**	0.008	*0.193*	*0.873*	*0.015*	**0.022**
German Credit (GC)	GC1	0.727	0.001	0.000	**0.727**	0.018	0.012	*0.560*	0.008	*0.103*	**0.727**	**0.001**	**0.000**	0.667	*0.050*	0.055
	GC2	0.747	0.014	0.000	0.727	0.011	0.036	0.693	**0.004**	0.035	**0.747**	0.014	**0.000**	*0.493*	*0.089*	*0.111*
	GC3	0.707	0.032	0.037	0.700	0.074	0.073	*0.320*	**0.032**	0.058	**0.707**	**0.032**	**0.037**	0.560	*0.166*	*0.196*
	GC4	0.713	0.015	0.000	**0.713**	0.015	**0.000**	0.707	0.010	0.036	0.707	*0.048*	*0.048*	*0.700*	**0.008**	**0.000**
	GC5	0.693	0.000	0.000	**0.693**	**0.000**	**0.000**	*0.307*	**0.000**	**0.000**	**0.693**	**0.000**	**0.000**	**0.693**	**0.000**	**0.000**
Adult Census (AC)	AC1	0.852	0.154	0.105	**0.847**	0.117	**0.031**	0.823	0.066	0.038	0.820	0.071	0.110	*0.773*	**0.028**	*0.165*
	AC2	0.847	0.119	0.028	**0.841**	0.090	0.036	0.831	0.071	**0.023**	0.825	0.071	0.105	*0.765*	**0.022**	*0.144*
	AC3	0.845	0.131	0.040	0.843	0.081	0.127	0.827	0.087	**0.013**	**0.844**	0.128	0.046	*0.776*	**0.025**	*0.181*
	AC4	0.849	0.134	0.066	**0.836**	0.087	0.041	0.828	0.072	**0.022**	0.816	0.053	*0.144*	*0.758*	**0.033**	0.119
	AC5	0.852	0.189	0.164	**0.848**	0.138	0.072	0.823	0.079	**0.002**	0.818	0.093	0.074	*0.789*	**0.029**	*0.190*
	AC6	0.848	0.131	0.077	**0.835**	0.085	**0.028**	0.822	0.062	0.029	0.811	0.046	0.152	*0.770*	**0.034**	*0.158*
	AC7	0.848	0.176	0.188	**0.844**	0.146	0.128	0.804	0.043	**0.056**	0.800	0.036	0.158	*0.778*	**0.027**	*0.160*
	AC8	0.828	0.263	0.178	**0.836**	0.235	*0.152*	0.787	0.085	**0.016**	0.809	0.146	0.060	*0.753*	**0.029**	0.136
	AC9	0.832	0.118	0.075	**0.818**	0.067	**0.041**	0.807	0.047	0.077	0.793	**0.013**	*0.191*	*0.735*	0.031	0.129
	AC10	0.845	0.204	0.159	**0.846**	0.143	0.063	0.804	0.074	**0.022**	0.815	0.102	0.081	*0.747*	**0.021**	*0.122*
	AC11	0.810	0.329	0.204	**0.838**	0.192	0.050	*0.751*	0.081	**0.027**	0.797	0.175	0.080	0.763	**0.045**	*0.118*
	AC12	0.842	0.181	0.104	**0.835**	0.087	0.068	0.812	0.076	**0.031**	0.813	0.071	0.147	*0.760*	**0.035**	*0.164*
COMPAS	Compas1	0.735	0.081	0.123	**0.730**	*0.194*	*0.309*	0.727	0.056	0.083	0.265	0.081	0.123	0.566	**0.000**	**0.000**
	Compas2	0.728	0.087	0.076	0.722	*0.190*	*0.262*	**0.727**	0.089	0.076	0.272	0.087	0.076	0.566	**0.000**	**0.000**
	Compas3	0.730	0.060	0.044	0.720	*0.215*	*0.323*	**0.728**	0.051	0.044	0.270	0.060	0.044	0.566	**0.000**	**0.000**
	Compas4	0.730	0.014	0.025	**0.725**	*0.230*	*0.351*	0.672	0.046	0.019	0.270	0.014	0.025	0.566	**0.000**	**0.000**
	Compas5	0.720	0.046	0.008	**0.714**	*0.167*	*0.202*	0.701	0.062	0.017	0.280	0.046	0.008	0.566	**0.000**	**0.000**
	Compas6	0.739	0.040	0.016	**0.733**	*0.195*	*0.271*	0.718	0.085	0.038	0.261	0.040	0.016	0.566	**0.000**	**0.000**
	Compas7	0.725	0.097	0.068	**0.717**	*0.158*	*0.207*	0.701	0.068	0.001	0.275	0.097	0.068	0.566	**0.000**	**0.000**

3.4.2. Multi-attribute Approach

We propose leveraging attribute pruning to understand more complex scenarios where multiple attributes are considered sensitive: for this approach, the weights in the first model layer for all the sensitive attributes are set to zero. To answer RQ2, we conduct a set of experiments on 24 models where 2 sensitive attributes are involved in the corresponding datasets. First, we prune the sensitive attributes and discuss the accuracy change before and after applying the approach. Then, we leverage an example to illustrate the procedure of measuring multi-attribute group fairness with true positive rate (TPR) difference based on multi-attribute pruning.

4. Experimental Results

4.1. RQ1: How does single-attribute pruning perform in comparison to the existing post-processing methods?

The performances of each model before and after applying the four single sensitive attribute post-processing methods are reported in Table 2.

Finding 1: Attribute pruning on small-scale DNNs will not severely harm accuracy. Instead, it obtained the highest accuracy on 23 out of 32 models among all post-processing methods. As analyzed in Sec. 2.3, pruning sensitive attributes will not severely decrease model accuracy. In contrast, it generally preserved the highest accuracies among the four compared methods. On six models (i.e., BM1, BM6, BM7, AC8, AC10, AC11), we even observed an increment in accuracy after pruning the sensitive attribute. The highest increment is observed on AC11, where the accuracy improved by 3.457%. The elimination of sensitive attributes prevented the model from overfitting to the training data on these models. On three models (i.e., GC1, GC4, GC5), the accuracy did not change after attribute pruning. The highest accuracy drop and its largest gap with the highest accuracy are observed on GC2, where the metric

Table 3
The upper bound of accuracy decrement and p-values for the paired t-tests on DP and EO, conducted on each dataset before and after the post-processing bias correction. For DP and EO, the p-values smaller than 0.05 (i.e., the distributions show significant differences) are highlighted in bold.

Dataset	Post-Processing Method											
	Attribute Pruning			EqOdds			CalibratedEqOdds			ROC		
	Acc	DP	EO	Acc	DP	EO	Acc	DP	EO	Acc	DP	EO
Bank Marketing (BM)	0.007	0.406	0.136	0.053	0.178	0.111	0.002	0.099	**<0.0001**	0.038	0.864	**<0.0001**
German Credit (GC)	0.029	0.530	0.079	0.957	0.848	0.068	0.012	0.563	0.478	0.540	0.143	0.124
Adult Census (AC)	0.034	**0.029**	**0.043**	0.066	**<0.0001**	**<0.0001**	0.063	**<0.0001**	0.880	0.117	**<0.0001**	0.093
COMPAS	0.013	**<0.0001**	**<0.0001**	0.085	0.727	0.554	0.502	≈ 1.000	≈ 1.000	0.180	**<0.0001**	**<0.0001**

decreased by only 2.677%. According to the one-sided normal tolerance interval test results displayed in Table 3, the accuracy decrement upper bounds before and after conducting attribute pruning are consistently less than 0.05 with a 95% coverage on 95% confidence level, suggesting that attribute pruning will not largely harm the model's performance.

On the other hand, other post-processing methods tend to have a more negative and unpredictable impact on accuracy: for instance, although the decrement upper bound calibrated EqOdds is 0.002 on the BM dataset, the value is 0.502 on the COMPAS dataset. Further, the ROC bias removal method obtained the worst accuracy on 24 models: this observation conflicts with the belief that ROC should balance fairness and accuracy. Especially when the ground truth positive rate is highly unbalanced: on the Adult Census dataset, ROC suffered from the largest accuracy drop on all models. On this dataset, male individuals having a positive ground truth almost tripled the female individuals in the same class. This can be due to historical bias against females, but may also be due to the limitation of the observed sample distribution (i.e., in this dataset, we do not observe as many qualified females as males, which is not aligned with the real-world distribution). Under the default settings provided in AIF360 (i.e., highest classification threshold used in the optimization is 0.99, the lowest threshold is 0.01, and the fairness optimization target is demographic parity), we found that ROC will perform a greedy search on the parameter space to force group fairness regardless of sampled individual quality difference among the two subgroups, which lead to impractical, overfitted decision margins (e.g., lowest threshold around 0.1 and highest threshold around 0.3). Such a decision margin obtained by greedy search then led to distorted accuracy. The other two methods, equalized odds and calibrated equalized odds, also tend to gain a lower accuracy than attribute pruning, even with TPR and FPR considered in their optimization goals. On the COMPAS dataset, we observed severe accuracy drops when the calibrated equalized odds method is applied for decision correction. All the original decisions are reversed by the algorithm (i.e., if the decision for an individual is positive, the corrected decision becomes negative after the correction) while preserving the same DP and EO value, suggesting that the algorithm converged to a trivial solution that flips labels but preserves the same group-level rates. The minimum drop is observed on Compas5, where the decrement is 61.111%, suggesting high unreliability in the modified decisions.

Finding 2: Attribute pruning may be able to improve group fairness, but the results are not usually optimal. Setting the sensitive attribute weight in the first layer to 0 should lead to identical predictions for similar individuals regardless of their sensitive attributes. By meeting this requirement, attribute pruning shall satisfy the strict individual fairness on the pruned sensitive attributes. However, according to Table 2, attribute pruning did not achieve the lowest DP or EO value on the majority of models. As also shown in Table 3, attribute pruning's impact on DP and EO is not significant on the Bank Marketing and German Credit datasets. Further, the results on the COMPAS dataset revealed that attribute pruning could significantly harm group fairness. This outcome reflected the conflict between individual fairness and group fairness stated by Binns [14]. This issue arises due to the datasets' distribution limitation. Group fairness could approximate individual fairness when the sample size is sufficiently large to cover the entire input space [7]. However, this is always impossible due to the existence of continuous or unlimited range features (e.g., working hours, loans). Due to the distribution incompletion of observed datasets, attribute pruning may fail to achieve the optimal group accuracies: this explains the increments in both DP and EO on GC1 and GC3, where the corresponding dataset only contains 1,000 samples. In the German Credit testing set, there are 119 privileged individuals but only 31 unprivileged individuals, and a single change in decision could lead to a large change in the two group accuracy metrics. The same explanation applies to the results on the COMPAS dataset: the dataset contains 2,103 privileged individuals and 4,069 unprivileged individuals; thus, the observations in the dataset are naturally unbalanced. We also observed that while the DP values generally decreased, all EO values on the 8 Bank Marketing

Table 4
The original accuracy of each model and the accuracy after multi-attribute pruning.

Dataset	German Credit (GC)					COMPAS						
Model	GC1	GC2	GC3	GC4	GC5	Compas1	Compas2	Compas3	Compas4	Compas5	Compas6	Compas7
Original	0.727	0.747	0.707	0.713	0.693	0.735	0.728	0.730	0.730	0.720	0.739	0.725
Attribute Pruning	0.733	0.707	0.673	0.713	0.693	0.722	0.718	0.712	0.718	0.718	0.717	0.710
Dataset	Adult Census (AC)											
Model	AC1	AC2	AC3	AC4	AC5	AC6	AC7	AC8	AC9	AC10	AC11	AC12
Original	0.852	0.847	0.845	0.849	0.852	0.848	0.848	0.828	0.832	0.845	0.810	0.842
Attribute Pruning	0.829	0.827	0.840	0.833	0.844	0.827	0.838	0.835	0.810	0.833	0.837	0.832

Table 5
AC3 and AC10's true positive rate (TPR) on different subgroups before and after multi-attribute pruning.

Model	Subgroup [sex, race]	[0, 0]	[0, 1]	[0, 2]	[0, 3]	[0, 4]	[1, 0]	[1, 1]	[1, 2]	[1, 3]	[1, 4]
AC3	Original TPR	0.941	0.914	0.953	0.944	0.903	0.925	0.746	0.872	0.905	0.808
	Corrected TPR	0.941	0.914	0.965	0.944	0.917	0.925	0.746	0.884	0.976	0.790
	$\|\Delta$TPR$\|$	0.000	0.000	0.012	0.000	0.014	0.000	0.000	0.012	0.071	0.018
AC10	Original TPR	0.971	0.886	0.959	1.000	0.916	0.950	0.705	0.863	0.929	0.803
	Corrected TPR	0.971	0.857	0.959	1.000	0.912	0.925	0.738	0.859	0.976	0.783
	$\|\Delta$TPR$\|$	0.000	0.029	0.000	0.000	0.004	0.025	0.033	0.004	0.047	0.020

models increased. However, this suggests that calibrating the decisions on the subgroup where $S = 1$ actually resulted in fewer false negatives for this group.

Conversely, the remaining three post-processing methods tend to obtain better group fairness in general, but some anti-patterns are also observed. The improvement on the German Credit dataset is not significant for all three methods according to Table 3. On the Adult Census dataset, although ROC achieved the lowest DP value on most of the models, its EO value is generally among the highest in the four methods. The reason also lies in the imbalance of ground truth positive rates between different subgroups: enforcing strong demographic parity harms the reliability of predictions in the observed distribution. Similarly, calibrated equalized odds received the highest EO and high DP values on the Bank Marketing dataset. On the COMPAS dataset, only ROC significantly improved the two fairness metrics: both metrics are reduced to 0.000, suggesting no group bias in the models. However, this approach significantly reduced the prediction accuracy from ≈ 0.730 to 0.566 because the algorithm greedily found a new threshold of 0.01 for all models, predicting every individual as belonging to the positive class. These outcomes suggest that optimizing specific group fairness on the training set may be able to enhance group fairness according to the corresponding criteria, but the improvement may not be significant, and it may fail to be generalizable to new data and other metrics. Consequently, they tend to be more "high risk in generalizability, high reward in certain fairness goal" in bias removal.

Summary: While ensuring individual fairness on the single attribute, attribute pruning will not significantly impact accuracy. It preserved the highest post-processing accuracy among the four methods on 23 out of 32 models. It can also improve the two group accuracies in general, but its improvements are not significant and not always optimal in comparison to the other three methods. Further, given the theoretical difference between individual fairness and group fairness, attribute pruning may even harm group fairness when the observed dataset is not comprehensive enough to cover the whole data space.

4.2. RQ2: How does multi-attribute pruning impact and aid understanding of the original models?

In comparison to single-attribute pruning, multi-attribute pruning has a stronger impact on accuracy, where 20 out of 24 models suffered from a slight accuracy drop. On the other hand, it still preserved competitive accuracy. The accuracy impact on multiple sensitive attributes pruning is shown in Table 4. The maximum decrease is observed on GC2 by 5.355%. However, 2 models, GC1 and AC11, obtained higher accuracies after disabling the two sensitive attributes. Another interesting discovery is that although the models suffered from a stronger accuracy decrement, 16 models still retained higher accuracies than the other three post-processing methods for fairness optimization on a single attribute.

Apart from further confirming the analysis in Sec.2.3, that sensitive attribute pruning will not lead to severe accuracy drop, another implication is **its potential usage for multi-attribute group fairness measurement in the**

original model. By disabling these features, the privileges and unprivileges assigned to individuals are fully eliminated. Therefore, we can further understand the attributes' impact on the model using their TPR differences (i.e., $|\Delta\text{TPR}|$) across subgroups. To illustrate the assessment process, we select two models, AC3 and AC10, which share the same original accuracy of 0.845 before attribute pruning. Their TPRs on different [sex, race] subgroups are shown in Table 5.

The sum of $|\Delta\text{TPR}|$ for AC3 is 0.127, while the value for AC-10 is 0.162. Hence, we say that the two sensitive attributes may have a larger impact on AC-10 than on AC3 in general. Apart from the general comparison, we can also gain insights into the models' decision impacts on specific subgroups. For instance, on subgroup [1, 3], it can be observed that the $|\Delta\text{TPR}|$ value of AC3 is 0.071, higher than AC10 (0.047), which means the original model tends to make more biased decisions against this subgroup of individuals. This information is useful for practitioners in choosing models to meet specific requirements (e.g., general fairness or protection on specific subgroups), especially when the models share the same accuracy. The comparison of subgroup $|\Delta\text{TPR}|$ is seemingly similar to the existing multi-group fairness metrics, such as ensuring the accuracy discrepancy on different subgroups is below a threshold, but we argue that this comparison tends to be more robust as it considers the underlying input distribution of different subgroups. We further elaborate on the discussion in Sec. 5.3.

Summary: According to our experiment on 24 models, multi-attribute pruning can also retain a certain level of accuracy while enhancing individual fairness. It can also be used to estimate multi-attribute group fairness in models with similar original accuracy based on the TPR difference before and after pruning the sensitive attributes.

5. Discussion

5.1. Approximating Individual Justice

As discussed in Sec. 2.3, attribute pruning ensures individual fairness by removing the direct contribution of sensitive attributes. However, it does not guarantee individual justice, which considers fairness beyond identical selected features, including removing biases that are indirectly captured through correlated non-sensitive attributes[14].

In this section, we discuss strategies to mitigate the impact of non-sensitive attributes that imply sensitive information, with the goal of approximating individual justice. We begin with one-on-one cases (i.e., one non-sensitive attribute indicates one sensitive attribute). There are two possible relationships between the two attributes; for each relationship, we provide an example to elaborate:

1. The non-sensitive attribute strongly predicts the sensitive attribute.
2. The correlation between the two attributes is seemingly weak or conditional.

Example 1: Assume a bank credit dataset for professors and undergraduate students in a university. The majority of professors are assumed to be older than undergraduate students, and the age attribute is regarded as sensitive to avoid discrimination. However, their professions are not anonymized, and the two attributes tend to share the same distribution. In this case, pruning the "non-sensitive" attribute may reduce this implicit bias.

Example 2: Consider postal codes as a proxy for race. While some postal codes are racially homogeneous, this correlation varies across regions. Therefore, this issue needs to be discussed according to domain knowledge on the dataset: if the correlation is common across most data, then it falls back to the first case, and pruning is advised; if it only affects a small subset, domain knowledge should guide whether to retain or remove the attribute.

The cases are more complex when multiple attributes are involved, since implicit biases may combine, and subtle correlations can remain even after pruning. For instance, we may remove postal code, gender, and date of birth to reduce re-identification risk [54], but features such as "alma mater" and "career" may also be encoded with sensitive information: graduating from a girls' school implies female sex, and working in construction is statistically more likely for males. Such biases are difficult to enumerate algorithmically and may impact only certain small subgroups. Furthermore, future shifts in data distribution can amplify these effects.

Therefore, attribute selection should be guided by both an understanding of historical data and anticipation of future data distributions to mitigate implicit bias. Additional techniques, such as adversarial debiasing or representation learning, can also help suppress bias. Nonetheless, the pursuit of full individual justice is inherently limited in machine learning systems, since their decisions are based on feature abstractions of individuals drawn from incomplete distributions. In cases where strict individual justice is required, as Binns notes, **human judgment remains preferable to algorithmic decision-making** [14].

5.2. Group Fairness Improvement with Attribute Pruning

Although attribute pruning is regarded as a post-processing method for improving individual fairness, we argue that it can also be used for group fairness improvement. As illustrated in Table 2, 3, and 4, attribute pruning will not affect the overall prediction accuracy. Instead, it calibrates the previous predictions by removing the impacts of the sensitive attributes. Hence, we can say that the decisions after attribute pruning are **as accurate as the original model, but fairer to individuals**.

Therefore, engineering the decisions based on the predictions after sensitive attribute pruning should be more favorable than changing the decisions directly on the original models, since the outcomes with their sensitive attribute pruned are more explainable. For instance, assume two candidates are from different sex subgroups, male and female. The original model assigned the male candidate a higher confidence of getting accepted than the female. However, the confidence difference may be due to their non-sensitive attributes, such as educational backgrounds, or due to discrimination based on sex. The reason cannot be well-addressed by directly observing the outcomes. In contrast, if such a circumstance happens after attribute pruning, the decision will represent solely the non-sensitive attributes, which is more explainable and socially acceptable to individuals. These more reliable values can be used to better adjust decisions to meet group fairness requirements.

5.3. The Benefit of Using TPR Difference in Multi-Attribute Fairness Estimation

Existing approaches on multi-attribute group fairness investigate and suppress the bias using methods such as multicalibration [8]. Multicalibration ensures that predicted probabilities align with actual outcomes across different subgroups, reducing systematic biases in confidence estimates. For example, a well-calibrated model is expected to reach the same prediction accuracy on different race subgroups, and its fairness is estimated by evaluating whether predicted probabilities are properly aligned with the ground truth labels. However, while this should be the ideal outcome across the entire input space, enforcing multicalibration on a limited dataset presents challenges.

The dataset used for model training and validation is always a subset and an approximation of the real-world distribution. A large dataset randomly collected can better model the real distribution, but it is still infeasible to enumerate all the possible combinations of features. Consequently, a model may make erroneous predictions for certain features because it has not learned their optimal representations. These features can be sensitive attributes, however, they can also be non-sensitive attributes. Therefore, a seemingly low accuracy on a subgroup may not certainly due to the discrimination on sensitive attributes, but due to the representation defect on certain non-sensitive attributes. The low accuracy may even be amplified due to the under-representation of subgroups (i.e., Inuits, American Indians), whose sample sizes are inherently small. These subgroups with less population may lead to poor probability estimations, and distribution shifts between training and real-world data can undermine fairness guarantees [14]. For example, greedily enforcing these strict fairness constraints may sometimes reduce overall model accuracy and decision reliability, as exhibited during our experiments on ROC bias removal in Table 2.

Instead, multi-attribute group fairness estimation with $|\Delta\text{TPR}|$ is more robust to subgroup distribution distortion, as the performance comparisons are done **within subgroups** rather than across the entire population. This comparison ensures that non-sensitive attributes are controlled within each subgroup, reducing confounding effects; the only factors affecting the decision change in a subgroup are sensitive attributes. By analyzing $|\Delta\text{TPR}|$, one can directly assess whether these attributes influence decisions in specific subgroups and quantify the extent of their impact. Other metrics, such as subgroup accuracy delta, can also be applied for fairness estimation in the same fashion. As mentioned previously in Sec. 4, this straightforward yet informative approach can provide highly useful insights for practitioners to understand the models' behavior and select models based on different fairness requirements.

5.4. Applying Attribute Pruning in Real-life Scenarios

To conduct post-processing attribute pruning in real-life scenarios, one shall first identify the sensitive feature in the dataset. As previously discussed in Sec. 2.3, such selection requires domain knowledge: some "conventionally" biased features are critical in some classification tasks, for example, the sex attribute in disease predictions; on the other hand, as mentioned in Sec. 5.1, some seemingly non-sensitive features may highly correlate with sensitive information. Therefore, the target attribute to be pruned should be examined by domain experts who have a solid understanding of the task.

After identifying the sensitive attributes, one can leverage this approach to improve model fairness or quantify the bias. Note that in Sec. 4.1, we showed that decisions ensuring individual fairness may conflict with group fairness requirements. Therefore, for tasks prioritizing group fairness, we suggest combining attribute pruning with other group fairness improvement approaches. We elaborated on the rationale in Sec. 5.2.

6. Threats to Validity

6.1. Construction Validity

In RQ1, we discussed the benefit of post-processing attribute pruning by comparing it with three state-of-the-art post-processing bias removal algorithms. Note that we excluded pre-processing and in-processing algorithms due to the extra training process requirement: the process introduces multiple variables, such as epoch, learning rate, random seeds, etc. Without a careful tuning process, the retrained model may provide a non-optimal result, which could lead to biased conclusions. To avoid these outcomes, we limit our comparison to post-processing approaches under default settings.

While our study highlights that post-processing attribute pruning preserves high accuracy, fairness, and accuracy trade-offs may depend on actual needs. For instance, in certain cases, the model is required to prioritize demographic parity over accuracy. Under these circumstances, it may not be ideal to solely leverage attribute pruning as a bias removal method. Instead, as discussed in Sec. 5.2, one should consider combining attribute pruning with other bias mitigation methods to provide decisions that meet group fairness requirements while preserving individual fairness to the largest extent.

6.2. Internal Validity

The benchmark DNN models we used were taken from prior studies [53, 46, 47] and are known as having high accuracies (i.e., (near-)optimal representations of features are learned). However, we have not yet tested the impact of attribute pruning on models with less optimal performance. As analyzed in Sec. 2.3, attributing pruning may not be applicable in these cases because the learned representations are not yet reliable.

Although sensitive attribute pruning ensures individual fairness by design, it does not guarantee individual justice. To avoid confusion, we explicitly addressed the difference between these two notions in Sec. 2.3 and provided suggestions for approximating individual justice by mitigating indirect discrimination in Sec. 5.1.

6.3. External Validity

An essential concern of external validity is the approach's generalizability. Although we did not experiment with this bias removal method on large machine learning models, its benefit on models such as GPT-2 has already been addressed in previous research [32]. Our work completed the missing piece of attribute pruning effectiveness evaluation on small-scale DNNs.

The datasets used in our study (i.e., Bank Marketing, Adult Census, German Credit, and COMPAS) are widely used in fairness research and cover different domains. However, they may not adequately reflect the real-world distribution. For example, the German Credit and COMPAS dataset contains fewer than 10,000 samples in total, which have a high chance of failing to represent the real population. On the other hand, the other two datasets are large in size, containing more than 40,000 samples, which ensures the generalizability of our findings. Future experiments can be done on other domains, such as healthcare, to further ensure the approach's applicability.

7. Conclusion

Our work proved the effectiveness of attribute pruning, an efficient, interpretable, and practical approach to post-processing individual fairness correction on small-scale DNNs. By directly pruning single sensitive attributes, we ensure individual fairness with minimal accuracy loss. However, due to the definition difference and inherent trade-offs between individual and group fairness, attribute pruning may fail to optimize group fairness. We thus suggest practitioners combine attribute pruning with other bias removal approaches when group fairness is the major focus.

Our experiments further show that this method can be extended to multiple sensitive attributes, enhancing its applicability to more complex scenarios. Additionally, its utility in bias measurement and model selection promotes greater transparency in fairness-aware AI development. Moreover, we explicitly discussed the approximation towards individual justice concerning non-sensitive attributes encoded with sensitive information, and proposed suggestions on meeting this fine-grained fairness requirement.

Given post-processing attribute pruning's effectiveness in individual fairness improvement and robustness in fairness estimation, we encourage future research to adopt this lightweight approach as a comparison baseline, since "less is more": simple methods can often prove highly effective.

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